

Cost Function for Linear Regression

Concepts, fixes, and visual insights from the lab notebook

1. Overview

This report explains the machine-learning concepts covered in the lab notebook “Cost Function” (Course 1, Week 1, Lab 3 of the Machine Learning Specialization), summarizes the issues found while preparing a clean, runnable version of the notebook, and walks through the visual insights produced by the corrected code.

The lab's goal is to build intuition for the cost function used in linear regression the quantity that measures how well a straight-line model fits a set of data points, and that gradient descent later tries to minimize.

2. What the Notebook Does

The notebook models the relationship between house size and house price with a simple linear function, and studies how well different choices of that function's parameters fit two small housing datasets.

2.1 The linear model

$$f_{w,b}(x) = w \cdot x + b$$

Here x is the input feature (house size, in units of 1000 sq ft), w is the slope (“weight”), and b is the intercept (“bias”). Together (w, b) define one specific straight-line model; different (w, b) pairs draw different lines through the data.

2.2 The cost function

$$J(w,b) = (1 / 2m) \cdot \sum [f_{w,b}(x^i) - y^i]^2, \text{ sum over } i = 0 \dots m-1$$

For each training example i , the model's prediction $f_{w,b}(x^i)$ is compared with the true target y^i . The difference is squared so it is always non-negative and larger errors are penalized more and then averaged over all m examples (the factor $1/2$ is a convention that simplifies the derivative used later for gradient descent).

Symbol	Meaning	Role
m	Number of training examples	Determines the averaging factor $1/2m$
w, b	Model parameters	What the learning algorithm searches over
$f_{w,b}(x)$	Model's prediction	Compared against the true label y
$J(w, b)$	Cost	Single number summarizing overall fit quality (lower = better)

3. Visual Insights

3.1 How cost changes as w varies

With b held at 100 (found in the earlier lab to be a good intercept for the two-point dataset), sweeping w shows two things at once: what the resulting line looks like against the data (left), and the resulting cost $J(w, 100)$ (right).

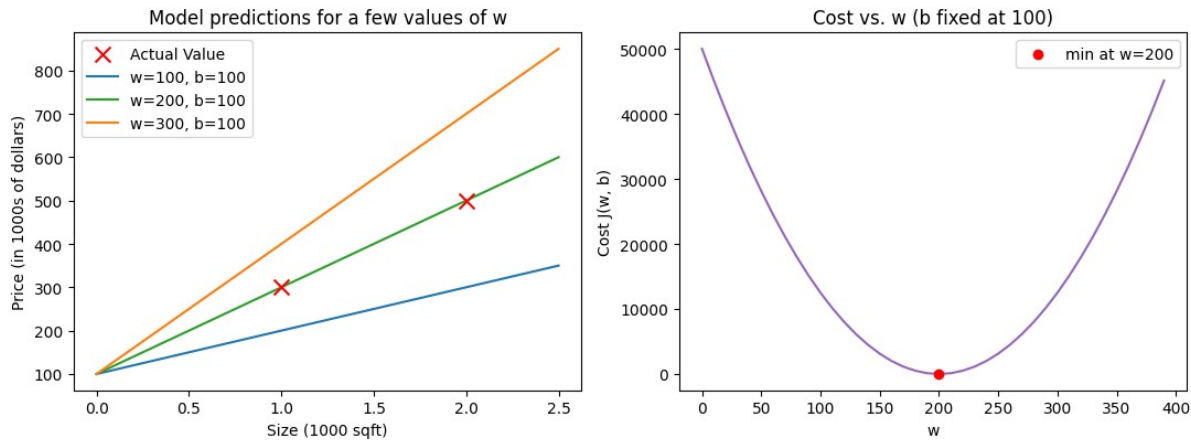


Figure 1. Candidate model lines for different w (left) and the corresponding cost curve (right).

The cost curve is a parabola in w , with its minimum exactly at $w = 200$ the value that makes the model's line pass through both data points exactly, driving cost to 0. Because the error term is squared, cost rises increasingly steeply as w moves away from its optimum in either direction; there is no flat region to get “stuck” in.

3.2 Cost as a function of both w and b

Once the dataset has more than two points that don't all sit on one line, no choice of (w, b) can drive the cost all the way to zero. The contour plot below evaluates $J(w, b)$ over a grid of both parameters at once for the larger, six-point dataset.

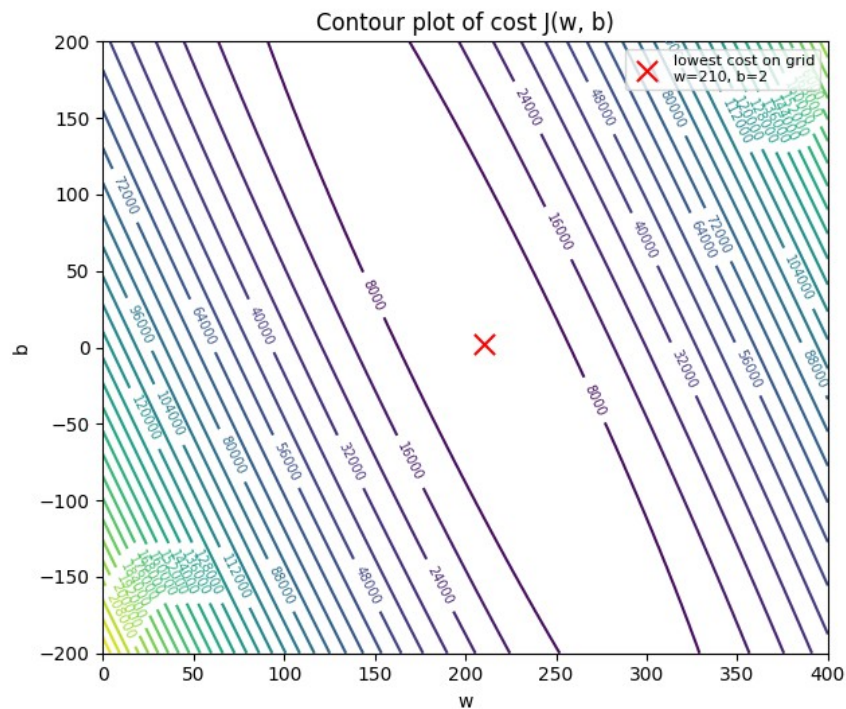


Figure 2. Contour plot of $J(w, b)$; each ring is a level curve of equal cost, and the marked point is the lowest cost found on the grid.

The concentric, roughly elliptical contours shrink toward a single low point near $w \approx 209$, $b \approx 2.4$, matching the value reported in the original lab. Moving away from that point in any direction increases cost there is only one “basin.”

3.3 The cost surface is convex

Plotting the same grid of $J(w, b)$ values as a 3D surface makes the shape behind the contour plot easy to see directly: a smooth, bowl-shaped surface with a single global minimum.

Cost surface is convex (a "soup bowl")

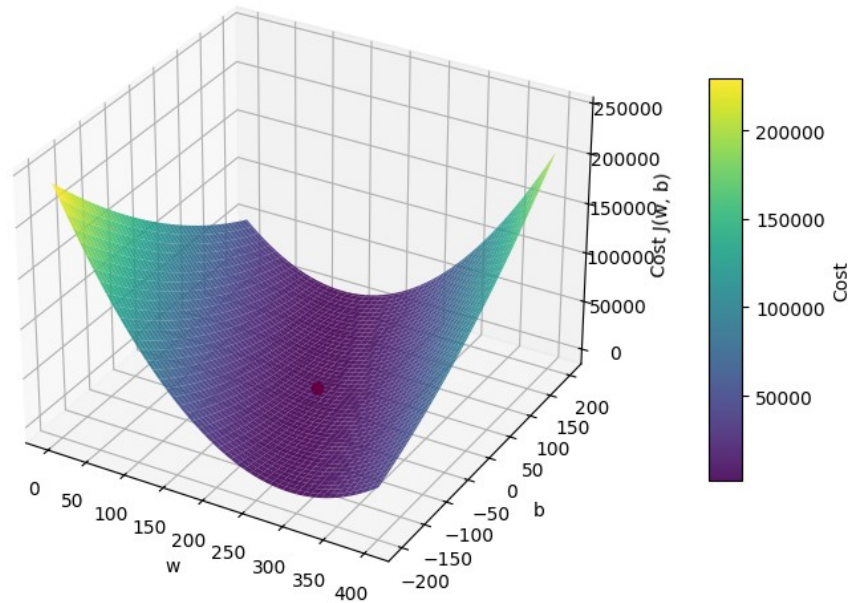


Figure 3. The squared-error cost surface for linear regression, shown over a grid of w and b .

This convexity is the key practical reason gradient descent works reliably for linear regression with a squared-error cost: from any starting point on the bowl, following the slope downhill always leads to the same single minimum there are no separate local minima to get trapped in.

4. Key Takeaways

- The cost function $J(w, b)$ is a single number summarizing how well a linear model fits the training data lower is better, and 0 means a perfect fit.
- Squaring the error makes the cost sensitive to large mistakes and produces a smooth, differentiable surface.
- For simple linear regression with squared error, that surface is convex (“bowl-shaped”) in (w, b) , guaranteeing a single global minimum the theoretical foundation for why gradient descent converges reliably in this setting.
- The notebook's core algorithmic code was correct; the only barriers to running it were missing external files and the interactive-widget backend, both resolved in the fixed notebook provided with this report.